

Early Stopping in Neural Networks

➤ Introduction:

- When training a neural network, you usually run for **many epochs** (passes through the entire dataset).
- But training too long can cause the model to **overfit** — it learns noise instead of the true pattern.
- **Early Stopping** automatically **halts training** when performance on a **validation set** stops improving.

➤ How It Works

The Process:

1. **Split your dataset** into:
 1. **Training set** → used for learning.
 2. **Validation set** → used for monitoring performance.
2. **Train the model** normally across many epochs.
3. After each epoch:
 1. Compute **validation loss** (or validation accuracy).
 2. If validation loss **improves**, save the model.
 3. If validation loss **stops improving**, start counting “patience” epochs.
 4. If no improvement for a certain number of epochs (called **patience**), **stop training**.

Goal:

Stop training **right before** the model starts to overfit.

➤ Visual Intuition

Epoch	Training Loss	Validation Loss	What Happens
1–20	↓ decreases	↓ decreases	Model learning well
21–25	↓ decreases	↔ stagnates	Validation no longer improving
26–30	↓ decreases	↑ increases	Overfitting begins
Stop Here			Best generalization point

➤ Key Parameters

Parameter	Meaning
monitor	Metric to track (e.g., val_loss, val_accuracy)
patience	Number of epochs to wait before stopping

mode	'min' or 'max' depending on metric
restore_best_weights	Whether to revert to best model weights

➤ Advantages

Prevents **overfitting**

Saves **time and resources**

Automatically selects the **best model**

No manual tuning needed for stopping epoch

➤ Disadvantages

Might **stop too early** if validation fluctuates (can fix with higher patience).

Requires a **validation set** (reduces training data slightly).

➤ When to Use

Use Early Stopping when:

- You observe training loss ↓ but validation loss ↑.
- You don't know the ideal number of epochs in advance.
- You want efficient, automated training without overfitting.

➤ Summary Table

Concept	Description
Purpose	Stop training when validation performance stops improving
Prevents	Overfitting
Based On	Validation loss or accuracy
Parameter	Patience (epochs to wait)
Best Practice	Use with <code>restore_best_weights=True</code>

➤ In Simple Words

Early stopping means:

“Train for as long as you're improving and stop when you start to forget what you've learned.”